MILITARY SPECIFICATION

GUM PREVENTIVE COMPOUND, GASOLINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and one grade of gasoline gum preventive compound, a mixture of antioxidant and metal deactivator to be added to gasoline to retard the formation of gum.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-T-425	- Tinplate (Electrolytic).
TT-T-548	- Toluene, Technical.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets, Preparation and the Submission of.
FED-STD-601	- Rubber, Sampling and Testing.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5166 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6850

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MIL-STD-105	- Sampling Procedures and Tables for Inapection by Attributes.
MIL-STD-290	- Packaging, Packing, and Marking of Petroleum and Related Products.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z129.1-1976 - Precautionary Labeling of Hazardous Industrial Chemicals.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 381 - Existent Gum in Fuels by Jet Evaporation.
D 525 - Oxidation Stability of Gasoline (Induction Period Method).
D 4057 - Manual Sampling of Petroleum and Petroleum Products.

(The test methods listed above are included in volumes 05.01 and 05.03 of the Annual Book of ASTM Standards and are available individually. Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The gasoline gum preventive compound shall be a mixture of antioxidant of the hindered phenol or phenylenediamine type and metal deactivator of the disalicylidene propylenediamine type dissolved in toluene. The compounded material shall provide test results equal to or better than the comparison formulation listed in 6.3 when tested as required by this specification.

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3.2 Performance requirements.

3.2.1 Corrosion. When tested in accordance with 4.5.1, the tinplate specimens shall show no staining, discoloration, or corrosion, and the gum preventive compound shall show no sedimentation or separation after standing for 168 hours at room temperature (20° C to 25° C).

3.2.2 Gasoline miscibility. The gasoline gum preventive compound shall be miscible with the reference fuel and shall show no sedimentation or separation after standing for 48 hours at 15° C, when added to the reference fuel in specified proportions in accordance with 4.5.2.

3.2.3 Initial oxidation stability. An increase in oxidation stability, as required by 3.2.3.1, 3.2.3.2, 3.2.4.1, 3.2.4.2, 3.2.5.1, and 3.2.5.2, shall be defined as an increase in the induction period as determined by ASTM D 525 and a decrease in the existent gum as determined by ASTM D 381.

3.2.3.1 Uncontaminated fuel. The gasoline gum preventive compound shall increase the oxidation stability of the uncontaminated reference fuel by an amount equal to or greater than that of the increase produced by the formulation in 6.3 when tested in accordance with 4.5.3.3.

3.2.3.2 Contaminated fuel. The gasoline gum preventive compound shall increase the oxidation stability of the contaminated reference fuel by an amount equal to or greater than that of the increase produced by the formulation in 6.3 after contamination with cyclohexanebutyric acid copper salt when tested in accordance with 4.5.3.4.

3.2.4 Ability to stabilize fuel in storage.

3.2.4.1 Uncontaminated fuel. The gasoline gum preventive compound shall increase the oxidation stability of the heated uncontaminated reference fuel by an amount equal to or greater than that of the increase produced by the formulation in 6.3 when tested in accordance with 4.5.4.1.

3.2.4.2 Contaminated fuel. The gasoline gum preventive compound shall increase the oxidation stability of the heated contaminated reference fuel by an amount equal to or greater than that of the increase produced by the formulation in 6.3 after contamination with cyclohexanebutyric acid copper salt when tested in accordance with 4.5.4.2.

3.2.5 Storage stability of compound.

3.2.5.1 Uncontaminated fuel. The heated gasoline gum preventive compound shall increase the oxidation stability of the uncontaminated reference fuel by an amount equal to or greater than that of the increase produced by the formulation in 6.3 when tested in accordance with 4.5.5.1.

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3.2.5.2 Contaminated fuel. The gasoline gum preventive compound after having been subjected to heat, shall increase the oxidation stability of the contaminated reference fuel by an amount equal to or greater than that of the increase produced by the formulation in 6.3 after contamination with cyclohexanebutyric acid copper salt when tested in accordance with 4.5.5.2.

3.3 Material safety data sheets. A material safety data sheet shall be prepared in accordance with FED-STD-313. Questions pertinent to the effect of gasoline gum preventive compound on the health of personnel when used for its intended purpose shall be referred to the appropriate medical service who will act as advisor to the acquiring activity (see 4.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of the gasoline gum preventive compound offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot.

4.3 Sampling (see 6.2).

4.3.1 For inspection of packaging. A random sample of units of product (see 4.3.1.1) shall be taken from each lot in accordance with procedures prescribed in MIL-STD-105, inspection level II and an AQL of 2.5 percent defective.

4.3.1.1 Unit of product. A completed pack prepared for shipment shall be considered a unit of product.

4.3.2 For testing. Unless otherwise specified in the contract or order, each lot shall be sampled in accordance with ASTM D 4057. The sample shall consist of at least one liter of material.

4.4 Inspection.

4.4.1 Examination of packaging. Samples selected in accordance with 4.3.1 shall be examined in compliance with MIL-STD-290 and section 5 herein in regard to fill, closure, sealing, leakage, packing and marking. Any unit in the sample

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having one or more defects shall be rejected and if the number of defective units equals or exceeds the rejection number for the appropriate sampling plan of MIL-STD-105, the lot shall be rejected.

4.4.2 Testing. The sample selected as in 4.3.2 shall be tested in accordance with the test procedures of 4.5. If the sample fails to comply with any requirement of section 3, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 Corrosion. A test strip made of tinplated steel conforming to QQ-T-425, Tinplate (Electrolytic), approximately 76 x 13 x 0.8 mm, shall be immersed in 15 mL of gasoline gum preventive compound contained in a stoppered test tube so that approximately 13 - 25 mm of the test strip is exposed to the air. After 168 hours contact at room temperature (20° C - 25° C), the strip shall be removed and wiped dry with a lint-free cloth and examined for evidence of staining, discoloration, and corrosion. The liquid shall be examined for homogeneity and sedimentation.

4.5.2 Gasoline miscibility.

4.5.2.1 Reference fuel. The composition of the reference fuel shall be as shown in table I. Where test procedures call for a reference fuel, it shall be in accordance with composition shown in table I.

TABLE I. Reference fuel.

Constituent	Property	Requirement
300 mL n-octane	Boiling point	124° - 126° C
300 mL 2,2,4-trimethylpentane	Purity	99.5% pure, min.
200 mL n-heptane	Boiling point	97° - 99° C
100 mL cyclohexene	Boiling point	82° - 84° C
100 mL mixture of 1-octene and 2-octene	Boiling point	124° - 126° C.

4.5.2.2 Procedure. Place 0.3 mL of gasoline gum preventive compound and 100 mL of reference fuel in a glass stoppered 100-mL graduated cylinder. Thoroughly mix by shaking and allow to stand for 48 hours in a constant temperature bath of 15° ±1° C. At the end of the test period the cylinder shall be removed from the bath, held against a white background and examined for evidence of sedimentation or separation.

4.5.3 Initial oxidation stability.

4.5.3.1 Copper stock solution. Dissolve 126.5 ±1.0 mg of cyclohexanebutyric acid copper salt in 1.00 liter of toluene conforming to TT-T-548. The resulting solution contains 20 mg/L of copper.

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4.5.3.2 Dilute gasoline gum preventive solution. Add one volume of gasoline gum preventive compound to nine volumes of toluene conforming to TT-T-548. The resulting solution contains 10 percent by volume of gasoline gum preventive compound.

4.5.3.3 Uncontaminated test fuel. Add 1.5 mL of the dilute gasoline gum preventive solution prepared in 4.5.3.2 to 50 mL of the reference fuel. Test the mixture for oxidation stability and existent gum in accordance with ASTM D 525 and ASTM D 381, respectively.

4.5.3.4 Contaminated test fuel. Add 0.50 mL of the copper stock solution prepared in 4.5.3.1 to 50 ML of the reference fuel. Add 1.5 mL of the dilute gasoline gum preventive solution prepared in 4.5.3.2. Test the mixture for oxidation stability and existent gum in accordance with ASTM D 525 and ASTM D381, respectively.

4.5.4 Ability to stabilize fuel in storage.

4.5.4.1 Uncontaminated test fuel. Add 1.5 mL of the dilute gasoline gum preventive solution prepared in 4.5.3.2 to 50 mL of the reference fuel in a tinplate friction closure can of 200 to 250 mL capacity conforming to PPP-C-96, type V, class 1 or 2. Cap the can tightly and heat in a bomb (see 6.4) at 71° C for 24 hours. Allow to cool to room temperature (20° C to 25° C). Test the mixture for oxidation stability and existent gum in accordance with ASTM D 525 and ASTM D 381 respectively.

4.5.4.2 Contaminated test fuel. Add 0.50 mL of copper stock solution prepared in 4.5.3.1 to 50 mL of the reference fuel in a tinplate friction closure can identical to the can used in 4.5.4.1. Add 1.5 mL of the dilute gasoline gum preventive solution prepared in 4.5.3.2, and cap tightly. Heat the can in a bomb (see 6.4) at 71° C for 24 hours. Allow to cool to room temperature (20° C to 25° C). Test the mixture for oxidation stability and existent gum in accordance with ASTM D 525 and ASTM D 381, respectively.

4.5.5 Storage stability of gasoline gum preventive compound.

4.5.5.1 Uncontaminated fuel. An unopened container of gasoline gum preventive compound shall be heated at 71° C for 48 hours and then allowed to cool to room temperature 20° C to 25° C). The material shall then be tested according to 4.5.3.3.

4.5.5.2 Contaminated test fuel. An unopened container of gasoline gum preventive compound shall be heated at 71° C for 48 hours and allowed to cool to room temperature (20° C to 25° C). The material shall then be tested according to 4.5.3.4.

4.6 Submission of material safety data sheets. The contractor shall furnish to the contracting activity the toxicological data and formulations required to evaluate the safety of the material for the proposed use through the submisssion of the material safety data sheet detailed in FED-STD-313.

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5. PACKAGING

5.1 Unit, intermediate and exterior packing and marking. Unit, intermediate and exterior packing and marking of the additive shall be in accordance with MIL-STD-290, level B, level C or commercial as specified (see 6.2). Type and size of unit container shall be as specified (see 6.2).

5.2 Additional marking. In addition to markings required by MIL-STD-290, each container shall be labeled in accordance with ANSI Z129.1-1976, page 10.

6. NOTES

6.1 Intended use. The gasoline gum preventive compound is intended to be added to gasoline to retard the formation of gum in small power units and vehicle fuel tanks during extended storage periods. It is to be added at the rate of 3 parts by volume of gasoline gum preventive compound to 1000 parts by volume of gasoline (1 ounce of gasoline gum preventive compound per 2.5 gal. gasoline). For use in the tropics it is to be added at the rate of 6 parts by volume of gasoline gum preventive compound to 1000 parts by volume of gasoline.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Quantity required in terms of U.S. gallons.
d. Sampling plan if different from that of section 4 of this specification.
e. Degree of packaging required (see 5.1).
f. Any special marking required (see 5.2).
g. Type and size of container (see 5.1).

The gasoline gum preventive compound will be purchased by volume, the unit being a U.S. gallon of 231 cubic inches (3.785 liters) at 60° F (15.6° C).

6.3 Formulation. The gasoline gum preventive compound is contemplated to be comparable in performance to the following approximate composition. However, the Government assumes no responsibility for the acceptance of a product claimed to be manufactured under the identical formula.

41.0 grams N,N'-disecondary butyl-para-phenylenediamine
10.3 grams N,N'-disalicylidene-1,2-propanediamine
3.78 liters toluene (toluol) conforming to TT-T-548

6.4 Oxygen bomb. An oxygen bomb with a 500-pound relief pressure gauge type such as described in method 7111 of FED-STD-601 is suitable for this purpose.

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6.5 Supersession data. This specification replaces VV-G-800, dated 19 February 1964 and MIL-G-14418, dated 29 October 1956, which were cancelled.

Custodians:
Army - ME
Navy - AS
Air Force - 68

Preparing activity:
Army - ME

Project 6850-0723

Review activities:
Army - GL, SM
Air Force - 11, 20

User activities:
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Navy - MC, SH, YD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-53042	Gum Preventive Compound, Gasoline

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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